

HARSH GUPTA

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| Portfolio |

EDUCATION

Vellore Institute of Technology, Bhopal

Bachelor of Technology in Computer Science (Artificial Intelligence & Machine Learning)

Sep 2023 – Present

CGPA: **9.10/10.0** (Top 5%)

Senior School Certificate Examination, CBSE

May 2022 85%

Secondary School Examination, CBSE

May 2020 80%

EXPERIENCE

AI/ML Intern | EnviroWealth

Dec 2025 – Feb 2026

- Built an AI-driven BRSR ESG pipeline using Python, FastAPI, and Gemini LLM to extract 400 plus compliance fields from PDFs.
- Designed a 3-layer NLP architecture with chunked Gemini prompting and a custom flat-to-nested JSON parser for structured output.
- Deployed Dockerized full-stack system on Vercel with MongoDB storage, cutting manual reporting effort by 70 percent end-to-end.

Software Engineering Intern | Mintzy

Nov 2025 – Jan 2026

- Built a Python FastAPI trading engine integrated into a 3-tier fintech architecture (Next.js → Node.js → FastAPI).
- Engineered advanced MongoDB data models for auth, plugin lifecycle, credit billing, and broker session management using Mongoose.
- Enforced JWT-based server-side identity extraction and credit-based rate limiting for a SEBI-compliant trading platform.

PROJECTS

Multi-PDF RAG Chatbot | Next.js 14, FastAPI, LangGraph, LangChain, Supabase, Groq, Gemini

[Live Demo]

- Engineered a full stack Multi PDF RAG chatbot with a 4 node LangGraph pipeline delivering real time streamed responses using Groq LLaMA 3.1 8B, with Google OAuth and JWT authenticated FastAPI endpoints.
- Engineered a user scoped vector ingestion pipeline using PyPDFLoader, RecursiveCharacterTextSplitter, Gemini embeddings, and Supabase pgvector, enabling semantic search across 1000 plus PDFs.
- Implemented user isolated retrieval with cosine similarity search and SSE based token streaming, improving response relevance and enabling low latency AI interactions.

Agentic Blog Writer | LangGraph, LangChain, Groq (LLaMA-3.3 70B), Streamlit, Tavily

[Live Demo]

- Architected a multi agent blog generation system using LangGraph state machines with 5 specialized nodes including Router, Researcher, Orchestrator, Worker, and Reducer, enabling autonomous content creation and reducing writing time by 80 percent.
- Developed a 3-mode research pipeline (Closed Book, Hybrid, Open Book) leveraging Tavily Search API with multi-query formulation and token-aware context truncation, ensuring factually grounded, citation-backed content across all topic categories.
- Launched a stateless Streamlit application on Community Cloud backed by Groq LLaMA-3.3 70B, featuring live graph execution streaming, real-time node progress tracking, and one-click Markdown export.

LLaMA-3 QLoRA Fine-Tuning | QLoRA, LoRA, Hugging Face, TRL, Accelerate, bitsandbytes

[Colab]

- Fine-tuned LLaMA-3 8B using QLoRA and LoRA adapter methods with Hugging Face Transformers, reducing GPU memory consumption by 50 percent while preserving full model accuracy and enabling training on consumer-grade hardware.
- Streamlined preprocessing, tokenization, and publishing of 2,000 plus structured JSON records to Hugging Face Datasets Hub, reducing manual data preparation time and enabling reproducible dataset versioning for downstream evaluation.
- Optimized the end-to-end training pipeline using TRL, Accelerate, and bitsandbytes with gradient checkpointing, AdamW-8bit optimizer, and mixed-precision training, improving efficiency by 30 percent and reducing peak VRAM usage significantly.

ACHIEVEMENTS

- Codeforces Pupil (Max Rating: 1313)** — Achieved a global rating of 1263 through consistent participation in competitive programming contests with strong algorithmic skills.
- IEEE Published Researcher** — 2 papers accepted at **IEEE ICTBIG 2025** (SUAS Indore) on ML-based diabetes risk prediction and multiclass cyberattack intrusion detection.
- Winner** — VIT Mother's Day Challenge; designed and shipped a live interactive web application driving engagement among 100 plus participants [Live Demo].
- First Runner Up** — Led a team during the Hack4Health Hackathon at VIT Bhopal, developing an AI based healthcare solution under competitive timelines.

TECHNICAL SKILLS

AI / ML

Scikit-learn, Pandas, NumPy, TensorFlow, Keras, Hugging Face, RAG, Generative AI, LLMs, CNNs, EfficientNet, XGBoost, LightGBM, OpenCV, Tesseract, EasyOCR, MobileNet, ResNet, VGG16

LLM / Agents

LangChain, LangGraph, LangSmith, llama.cpp, LlamaIndex, QLoRA, LoRA, Unsloth, Transformers, PEFT, TRL

Programming

Python (DSA), Java, C++, TypeScript, JavaScript, HTML, CSS, SQL

Backend

FastAPI, Node.js, Next.js 14, REST APIs, Server-Sent Events, JWT, Supabase, pgvector

Cloud/DevOps

Docker, Git, GitHub, Flask, Streamlit, MongoDB, Render, Vercel, Postman

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate**
- Generative AI using IBM Watsonx** — IBM Developer Skill Network
- MATLAB and Simulink Technical Computing Certification** — MathWorks
- GeeksforGeeks 160-Day DSA Mastery Certificate** — GeeksforGeeks

Jul 2025

Jun 2025